# 2025-spring Introduction to Quantum Information and Quantum Computing

HW # 2 (Due 4/10 midnight = 4/11 00:00 am). Hand in as a single pdf file through ETL. For Qiskit problems: Pasting only the result (output) of the code (perhaps as a picture) will suffice—no need to hand in the coding itself unless explicitly specified. Also, running on an actual backend is recommended if possible, but the simulation result is also Okay.

1. Do "exercise 4.9: Schmidt decomposition and partial trace" of the lecture note.
2. **W state.** Is an entangled 3-qubit state given by $|W\rangle = \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle)$. Calculate entanglement measure for this state and compare with that of 3-qubit GHZ state.
3. **Qiskit –** Search internet for quantum circuit generating 3,4, and 5 qubit W states. Implement the circuit by qiskit and perform quantum state tomography. If possible, calculate state fidelities for each case.
4. **Qiskit –** Do DIYs of the "Quantum teleportation with Qiskit" file. Try your best to explicitly show your reasoning.
5. **Learn by searching** – Search with keyword "Quantum repeater" and related ones, write a few paragraphs explaining its working principle and current experimental challenges of building one. You can get aid from AI tools, but do not just copy & paste. Write your own paragraphs.
6. **Qiskit –** Do DIYs of the "Simon Algorithm with Qiskit" file. Try your best to explicitly show your reasoning. For the last one (#3), paste your code.